



Voice spoofing detection for multiclass attack classification using deep learning

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ABSTRACT

Voice biometric authentication is increasingly gaining adoption in organisations with high-volume identity verifications and for providing access to physical and other virtual spaces. In this form of authentication, the user's identity is verified with their voice. However, these systems are susceptible to voice spoofing attacks as malicious actors employ different types of attacks such as speech synthesis, voice conversion or imitations, and recorded replays to spoof the Automatic Speaker Verification (ASV) system or for spam communications. In this work, we provide a voice spoofing countermeasure as a binary classification problem, that classifies real and fake audio, and also as a multiclass classification problem to detect voice conversion, synthesis and replay attacks. We investigated numerous audio features and examined each feature capability alongside state-of-the-art deep learning algorithms including convolutional neural networks (CNN), WaveNet, and recurrent neural network variants — Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM) models. Using a large dataset of 419,426 audio files for experiments, we evaluated the deep learning models for their effectiveness against voice spoofing attacks. The binary class CNN achieved a false positive rate (FPR) of 0.0216, while the multiclass solutions using CNN, WaveNet, LSTMs and GRUs achieved an FPR of 0.003, 0.0260, 0.0302 and 0.0358 respectively. We extended the evaluation of the models by including the real-time classification using microphone voice audio and user-uploaded audio to demonstrate the practical implications and deployability.

1. Introduction

Voice phishing attacks and scam phone call detection have seen multiple approaches which are mainly utilising tracking call detail records (CDRs) and analysing generated transcripts. However, this approach is inhibited because (i) the availability of an open dataset of voice phishing or phone call logs (ii) the volume of new unseen records that come regularly (iii) user privacy intrusion limits the effectiveness of analysing call transcripts. (iv) CDRs can be used to assess whether a call is fraudulent or not, but the actual audio content is overlooked (Elizalde & Emmanouilidou, 2021). As a result, in the context of voice, approaches have focused on voice spoofing, which can be part of the phishing process. The audio content analysis comes with the additional benefit of privacy protection because it does not investigate the spoken content or transcript. Voice spoofing is the use of technology to alter, create, or imitate authentic voice (truly spoken by an actual human) (Wu, Evans et al., 2015). Voice spoofing attacks have increased as fraudsters, intruders and spammers employ different types of attacks such as speech synthesis, voice conversion or imitation, and recorded replay to try to spoof systems or for spam communications. Given that the *real voice* is the authentic spoken audio that has been truly spoken

by a human, the *voice conversion* attack involves manipulating voice audio to make it sound like that of another person, while maintaining the spoken content conveyed by the voice audio (Sisman, Yamagishi, King, & Li, 2020). The *synthesised voice* attack involves artificially generating voice audio to imitate human speech (Allen, Hunnicutt, Klatt, Armstrong, & Pisoni, 1987). In the *voice replay* attacks, a real voice is recorded using a microphone, and the real voice audio is replayed through a speaker to make it seem like the replayed voice audio is spoken audio (Wu, Evans et al., 2015). The voice replay attacks can be single-order (replayed once) and multi-order (i.e. replayed twice or more).

Voice spoofing attacks are becoming commonplace as technology continues to play a larger part in our everyday lives. From software implementations such as biometric authentication in software and devices (Khoury, El Shafey, McCool, Günther, & Marcel, 2014), voice commands given to Internet of Things devices like the Amazon Echo (Malik, Malik, & Baumann, 2019), and the ability of voice to reach people around the globe through social media applications (Stahie, 2022). These are all avenues that are open to exploitation by malicious actors, and voice spoofing through conversion, synthesis and replay are exploitable attack vectors (Alegre, Amehraye, & Evans, 2013; Kinnunen

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et al., 2017; Perrot, Aversano, & Chollet, 2007; Stupp, 2019). Recorded voice audio can be used in replay attacks to biometrically authenticate access to accounts or physical properties through automated voice recognition systems (Ergünay, Khoury, Lazaridis, & Marcel, 2015), or to take command of internet-connected devices like smart home systems (Malik et al., 2019). Voice synthesis of the voice of high-profile people such as CEOs of organisations has contributed to \$1.7 billion in losses in 2019 alone by fooling unsuspecting victims of payment fraud into thinking that they are hearing genuine voices (Bateman, 2020). Real-time conversion of voice can be used to disguise the true voice of the speaker, such as over 50 billion spam calls were recorded in the USA in 2022 (Notaney, 2023). A journalist used an AI synthesis of his voice to bypass the voice authentication system for Lloyd's bank and access his account (Cox, 2023). One of the primary metrics Android employs to assess the security performance of a biometric system is the *spoof acceptance rate* (SAR), which indicates the likelihood that the security system may be bypassed with a fake credential (Security, 2023). Spoofing is thus a significant issue for authentication systems (Biggio, Akhtar, Fumera, Marcialis, & Roli, 2012; Chen, Trappe, & Martin, 2007; Hautamäki, Kinnunen, Hautamäki, Leino, & Laukkanen, 2013; Hu & Wang, 2018; Shoukry, Martin, Yona, Diggavi, & Srivastava, 2015; Tippenhauer, Pöpper, Rasmussen, & Capkun, 2011).

To provide countermeasures to voice spoofing attacks, several research efforts have been put into making use of artificial intelligence models to classify if spoken audio is real or fake (Ballesteros, Rodriguez-Ortega, Renza, & Arce, 2021; Baumann et al., 2021; Delgado et al., 2018; Gong, Yang, Huber, MacKnight, & Poellabauer, 2019; Hassan & Javed, 2021; Zhang, Jiang, & Duan, 2021; Zhou et al., 2022). However, these solutions considered the problem as a binary classification where the challenge is detecting whether audio is spoofed or not or considers only one attack type, thereby overlooking the sophistication of the attack type. As a result of formulating the problem as a binary classification of real and synthetic speech, real and replay speech or real and converted speech, existing methods suffer from generalisation to unseen spoofing attacks in the testing stage (Kamble, Sailor, Patil, & Li, 2020; Wu, Evans et al., 2015; Zhang et al., 2021). Furthermore, the solutions proposed in the state-of-the-art are limited in terms of the robustness of the detection corpus, acoustic features and algorithms investigated. Finally, the practical implications of the solutions are not evaluated in terms of deployability and actionability. The proposed solution aims to create and use a variety of deep learning classifiers to classify audio into the 'real', 'converted', 'synthesised', and 'replay' categories to combat the problem of voice spoofing by identifying fake from real audio. We propose a state-of-the-art voice spoofing countermeasure based on six core features which are the robustness of the dataset, feature evaluation, diverse deep learning algorithms, binary & multiclass classification, real-time classification and model deployment testbed. This technology contributes to the creation and improvement of systems such as automated speaker verification systems in Rahmeni, Aicha, and Ayed (2022).

This paper builds the current efforts into voice spoofing countermeasures using deep learning classifiers (Zhou et al., 2022) and makes the following unique contributions:

- **Robust Dataset:** Utilisation of a larger, collated dataset as opposed to a singular smaller dataset. The collated dataset contains *real*, *synthesised*, *replayed*, and *converted* voice audio files. The dataset represents a diverse voice spoofing detection corpus in terms of environments, attack types, microphones, recording and playback devices, replay scenarios, configurations, synthesis algorithms, sampling rate, conversion techniques etc. With a robust dataset, the performance of voice spoofing countermeasures was evaluated under different classes of attacks and diverse conditions.
- **Feature Evaluation:** Evaluation of spectral features offered by Librosa (McFee et al., 2015) for binary and multiclass classification problems in the topic of voice spoofing. This is useful for identifying relevant and significant features for the classification tasks and informing future research about promising audio features.
- **Diverse Deep Learning Algorithms:** Implementation and evaluation of the suitability of multiple deep learning models such as CNNs, WaveNet, GRU and LSTM for their effectiveness against voice spoofing attacks as opposed to a single implementation of machine or deep learning models. Since this model uses different types of inputs ranging from audio graphs to raw audio, insights were provided on the best scenarios for each of the algorithms. Also, the evaluation of a variety of deep learning algorithms in voice audio classification helps in identifying which algorithms are best suited to address the problems associated with voice spoofing.
- **Binary and Multiclass Classification:** We investigated the voice spoofing detection problem as a binary and a multiclass problem. This involves the creation of binary models capable of classifying real and fake audio, and multi-class models capable of classifying real, synthesised, converted, and replayed audio as opposed to just binary models which classify between real or synthesised as in Ballesteros et al. (2021) and Zhang et al. (2021) or models that considers a single class of attack as in Baumann et al. (2021), Delgado et al. (2018) and Gong et al. (2019). With the binary classification model, the models are capable of predicting if audio is real or fake so that audio inputs can be identified as authentic or spoofed regardless of the attack vector. With the multiclass classification, the models are capable of predicting if audio is real, or if not, which attack vector is being exploited in the spoofed audio so that audio inputs can be identified as authentic, or the attack vector (converted, synthesised, or replayed) can be identified.
- **Real-Time Classification:** Evaluating the performance of the models on real-time classification from a microphone as opposed to pre-recorded audio files. The binary and multi-class models were evaluated to predict real-time audio inputs to investigate the performance of the models on novel audio data for real-life application of the models.
- **Model Deployment Testbed:** Deployment of optimal models to a web service so that users can easily upload audio files for classification while detecting voice spoofing attacks by informing the users if the audio they submit is authentic or not.

The rest of the paper is structured as follows. Section 2 provides a review of state-of-the-art voice spoofing countermeasures using deep learning. In Section 3, we highlight the development phases for the voice spoofing detection for multiclass attack classification using deep learning, the dataset collection process, the pipeline of the deep learning algorithms investigated in the study, audio feature extraction techniques, normalisation of audio data and rationale for the choice of methodology. In Section 4, we describe the research questions, experiments, results and discussion to validate the contribution of the proposed approach. We conclude in Section 5 presenting the summary of the research work and future directions.

2. Literature review

As summarised in Table 1, there are several limitations to the current research on voice spoofing countermeasures. Firstly, several of the current research papers such as Hassan and Javed (2021), Zhang et al. (2021) and Zhou et al. (2022), typically explore only binary classification of real and one fake attack vector class such as synthesis, replay, or conversion. As a result, some existing models may be unable to differentiate between the attack vectors employed in spoofed audio. Moreover, the dataset used by existing literature such

Table 1
Summary of existing research.

Ref.	Classification	Algorithm	Feature	Dataset
Baumann et al. (2021)	Multiclass: Real, First-degree replay & Second-degree replay attacks.	Gaussian Mixture Models (GMM)	Constant-Q Cepstral Coefficients (CQCC)	Real, first-degree & second-degree replay audio
Delgado et al. (2018)	Binary: Real & Replay	GMM	CQCC, Mel-Frequency Cepstral Coefficient (MFCC)	ASVSpooof2017 (Kinnunen et al., 2017) Contains real & replay audio
Gong et al. (2019)	Binary: Real & Replay	GMM	CQCC	Proprietary ReMASC Corpus (Gong et al., 2019) Containing real and replay audio
Ballesteros et al. (2021)	Binary: Real & Synthesised	CNN	Histogram, Spectrograms	Proprietary dataset of real and synthesised audio generated using deep voice and imitation
Hassan and Javed (2021)	Binary: Bonafide & Spoof where spoof is converted and synthesised audio	BiLSTM	MFCC, Spectral Centroid, Gammatone Cepstral Coefficient (GTCC), Spectral Flux	ASVSpooof2019 (Wang et al., 2020)
Arif, Javed, Alhameed, Jeribi, and Tahir (2021)	Binary: Bonafide & Spoof where spoof is converted and synthesised audio	DBiLSTM	ELTP, LFCC	ASVSpooof2019 (Wang et al., 2020)
Zhou et al. (2022)	Binary: Real & Replay	BiLSTM	GTCC, MFCC	ASVSpooof2019 (Wang et al., 2020)
Zhang et al. (2021)	Binary: Real & Synthesised	LCNN, ResNet	Feature Learning: LFCC	ASVSpooof2019 (Wang et al., 2020)
Chen, Xie, Zhang, and Xu (2017)	Binary: Real & Replay	GMM, RNN, DNN	MFCC, CQCC	ASVSpooof2017 (Kinnunen et al., 2017)

as Zhang et al. (2021) which only explores the logical access data contained in ASVSpooof2019 (Wang et al., 2020) is not extensive enough to represent voice spoofing attacks in voice-controlled systems. This can be because previous works focus on binary classification, but larger datasets are preferable, particularly in deep learning models for voice recognition tasks (Çayır & Navruz, 2021). In addition, some papers may only make use of singular algorithms for creating models such as Gong et al. (2019), Hassan and Javed (2021) and Zhou et al. (2022). This fails to fully explore the potential of a variety of models trained on the same data and lacks a comparison of the performance of each model. Some of the papers summarised in Table 1 such as Baumann et al. (2021), Delgado et al. (2018) and Zhou et al. (2022) tend to only explore a small number of audio features. The limitation of the small feature space is that it could result in failing to represent all the aspects of the audio data, and important features of the data may be left unexplored. This could limit the correctness of the classification predictions. Moreover, literature works typically make use of already researched and widely used features such as CQCC and MFCC (Delgado et al., 2018), so more obscure features that could prove useful in voice spoofing countermeasures may be missed.

In this paper, we address the limitations of existing research in several ways. Voice spoofing detection is formulated as a binary and multiclass classification problem, so the limitation of only exploring a binary classification problem is addressed. This contributes to the ability of the models in this research to classify audio as either *real* or *fake*, *bonafide* or *spoofed*, as well as the ability for models to identify the attack vector used by spoofed audio. Using a collated dataset of audio files containing *real*, *synthesised*, *converted* and *first* and *second-degree replay* audio, there is a larger availability of training, test and validation data for the deep learning models. This is beneficial as this research exclusively makes use of deep learning approaches such as CNNs, WaveNet, and RNNs (GRU and LSTM) which typically require larger datasets to achieve the best performance. Additionally, by utilising these algorithms in particular, each model contributes a unique perspective on the dataset. CNNs explore the significance of the features in image format, RNNs are trained on features in the format of NumPy arrays, and WaveNet is fed the raw audio input without any specifically extracted features. Feature correlation to the audio classification process is extensively researched, with the features offered by the Librosa Python library used and evaluated in their significance to model performance. As a result, a larger variety of features are researched across a variety of deep-learning algorithms in this paper.

3. Methodology

A generic framework for the proposed voice spoofing detection and multiclass attack classification using deep learning is presented in Fig. 1. The framework is divided into five development phases which include — the collation of datasets, extraction of spectral features, creation of deep learning models, evaluation of the created models, improvement of models through hyperparameter tuning, and model deployment of a front-end service for classification of audio files to combat voice spoofing by allowing users to determine if audio is real or fake, and the voice spoofing attack employed by fake audio. We discuss the various developmental phases below.

3.1. Dataset collection

The dataset used for evaluation consists of a collated dataset that has been used as a benchmark in literature. The reason behind choosing to make use of a collated dataset is that most public datasets typically only contain a binary combination of real and fake forms of voice audio. As this paper aims to combat voice spoofing of replayed, synthesised, and converted audio it was necessary to collate datasets so that sufficient audio files for each attack type and real voice files were available for training models. By making use of a variety of datasets and collating them into a singular larger dataset, this paper makes use of a diverse set of audio files. In particular, each dataset makes use of a range of hardware devices such as microphones to record real spoken audio or replay devices like phones or speakers in the replay attack audio. Additionally, in the case of synthesised and converted files, several synthesis algorithms and conversion techniques are employed. This makes the dataset more robust and therefore benefits each model's ability to classify audio in real-life use cases where an assortment of spoofing techniques may be used.

For example, existing datasets, such as ASVspooof2017 (Delgado et al., 2018; Kinnunen et al., 2017), ASVspooof 2019 (Wang et al., 2020), and ReMASC (Gong et al., 2019), for instance, only contain first-order replay recordings (i.e., repeated once), hence they are unable to provide an evaluation of anti-spoofing algorithms capable of identifying multi-order replay attacks. Replay attacks were not taken into consideration in several situations, such as ASVSpooof2015 (Wu, Kinnunen et al., 2015). Furthermore, it has been demonstrated in Baumann et al. (2021) that these datasets do not adequately represent the characteristics of microphone arrays, which are crucial features of modern voice-controlled devices. Therefore, there exists a need for a solution that consists of a diverse voice spoofing detection corpus that consists of

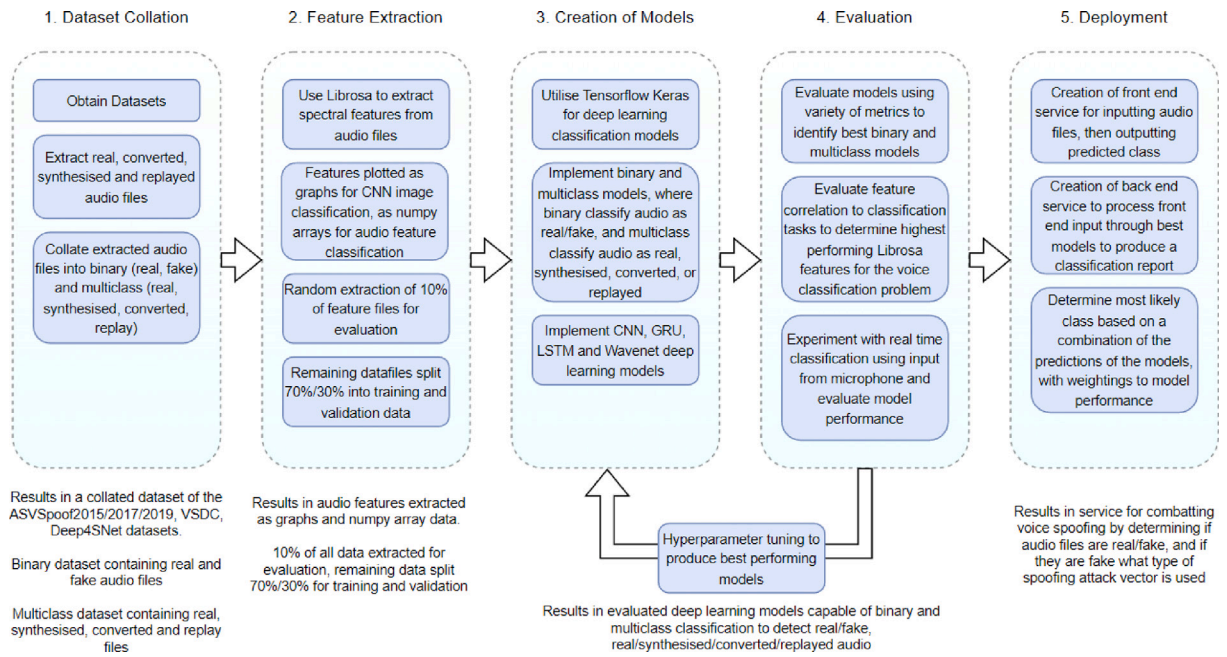


Fig. 1. A framework for voice spoofing detection and multiclass attack classification using deep learning.

Table 2
Diversity of collated dataset.

Dataset	Speakers	Recording devices	Recording microphones	Playback devices	Environments
VSDC	19	31	35	10	15
ASVSpoo2019	107	40	40	40	27
ReMASC	55	2	2	4	4
ASVSpoo2017	42	25	25	26	22
ASVSpoo2015	106	Not provided	Not provided	Not provided	Not provided

single and multi-order replay, synthesised and converted recordings against bonafide voice samples. The diversity of the detection corpus in terms of the number of speakers, recording devices, recording microphones, playback devices and environments is demonstrated in Table 2.

The breakdown of the number of files contributed by each of the datasets for different classes of attacks is shown in Table 3. Five publicly available datasets extensively used in literature for voice spoofing countermeasures were obtained and used in this research as described below:

- **The ASVSpoo2015** (Wu, Kinnunen et al., 2015) dataset contains audio files for *real*, *synthesised*, and *converted* audio. For the synthesis of audio files, Gaussian Mixture Model (GMM) and Hidden Markov Model (HMM) -based speech synthesis tools were used, with synthesis based on both 20 and 40 original utterances. For the conversion of audio, naïve frame-selection conversion was used to manipulate the original audio, and voice conversion through manipulation of the first Mel-Cepstral coefficient was also used.
- **ASVSpoo2017** (Delgado et al., 2018; Kinnunen et al., 2017) contributes *replay* attack audio data produced in an array of suitable and unsuitable environments such as indoors and outdoors. Additionally, many playback devices were employed, using hardware like PCs, phones, tablets, and headset speakers. Moreover, for recording *bonafide* audio a similarly diverse range of microphones was used with varying mobile phone microphones, tablets, webcams, and desktop microphones. Both *first-* and *second-degree replay* attacks are included in this dataset.
- **ASVSpoo2019** (Wang et al., 2020) is a robust dataset which contributes *real*, *synthesised*, *converted*, and *replayed* audio files.

Table 3
Dataset distribution.

Dataset	Real	Synthesised	Converted	Replay
ASVSpoo2015	4953	20 058	30 020	0
ASVSpoo2017	2267	0	0	2457
ASVSpoo2019	41 373	79 204	29 774	189 540
Deep4SNet	50	50	0	0
VSDC	2366	0	0	17 314
Total	51 009	99 312	59 794	209 310

- **Deep4SNet** (Ballesteros et al., 2021) contributes *real* audio files and *synthesised* recordings obtained using Deep Voice and Imitation.
- **Voice Spoofing Detection Corpus (VSDC)** (Baumann et al., 2021) is a dataset of audio files that include *bonafide*, *single* and *multi-order* audio replays.

Following the collation of the datasets described in Table 3, there is a clear data imbalance between the real, synthesised, converted, and replayed audio classes which could skew some evaluative metrics when evaluating the deep learning models. This data imbalance is expected due to the nature of voice spoofing where a single real voice can be spoofed many times in many ways. For example, a single real voice audio file may be used to generate multiple synthesised or converted voice audio files using different algorithms and software and could also be replayed through several device speakers. In other words, a single real voice file can result in the creation of many spoofed audio files so a data imbalance is inevitable in a robust dataset. To account for the data imbalance in model evaluation, imbalance-insensitive evaluative

metrics are used so that a true representation of model performance is reported irrespective of the class imbalance present.

The ASVspoof2019 (Wang et al., 2020) and Voice Spoofing Detection Corpus (VSDC) (Baumann et al., 2021) datasets contain *replay* audio which is classified as *first-degree* (recorded from *bonafide* audio), or *second-degree* (recording of replayed audio). However, it is important to note that in our approach, we treat *first-* and *second-degree replay* audio as a single replay class and do not aim to explicitly differentiate between the degree of a replay attack. As a result, when collating the datasets the degree of the replay audio was not considered and both first- and second-degree replay audio was bundled into a singular replay class. 10% of the audio files in the collated dataset were programmatically chosen at random for validation data, with the remaining 90% then being split 70/30 for training and test data. A 70/30 train/test split was chosen because, with more test samples, a better representation of initial model performance would be revealed (Nguyen et al., 2021). This allows better comparison between the algorithms used and the models fitted for every feature.

3.2. Deep learning model

The proposed approach in Fig. 1 uses deep learning algorithms rather than any traditional machine learning models as deep learning implementations are better at voice spoofing detection (Alam, Kenny, Gupta, & Stafylakis, 2016; Alzantot, Wang, & Srivastava, 2019; Arif et al., 2021; Ballesteros et al., 2021; Chen, Qian, Dinkel, Chen, & Yu, 2015; Chen et al., 2017; Delgado et al., 2018; Dinkel, Qian, & Yu, 2018; Gomez-Alanis, Gonzalez-Lopez, & Peinado, 2020; Gomez-Alanis, Peinado, Gonzalez, & Gomez, 2019; Gong et al., 2019; Lai, Chen, Villalba, & Dehak, 2019; Qian, Chen, & Yu, 2016; Wang & Su, 2019; Xiao et al., 2015). The suitability of different deep learning algorithms were investigated for voice spoofing countermeasures. The selected deep learning algorithms for developing predictive models used in this paper for voice spoofing detection for multiclass attack classification are described as follows:

- **Convolutional Neural Networks (CNNs):** CNNs as a deep learning network are suitable for image classification problems (Rawat & Wang, 2017). They generally consist of convolutions, fully connected, and pooling layers. They have been utilised for the classification of audio histogram graphs in Ballesteros et al. (2021), where the models were evaluated in the binary classification of synthesised and real voice audio. Inspired by these findings, and to contribute further to investigate the performance of audio features formatted as graphs, this paper aims to utilise CNNs for both binary and multiclass classification.
- **WaveNet:** WaveNet is a deep learning model created and used for synthesising raw audio in text-to-speech applications (Oord et al., 2016). It is comprised of dilated convolutional layers and gated activation units. This is beneficial as it increases the ability of the model to learn more complex features in the audio data. WaveNet has also been adapted for use in audio classification purposes. In Zhang, Gao, Yu, and Li (2020), the WaveNet classifier was suitable for the classification of music artists from audio inputs. As WaveNet has been demonstrated to perform well in the classification of audio in large datasets, we investigated the use of the WaveNet classification model produced in Zhang et al. (2020) to evaluate its performance in classifying raw voice audio data in the binary and multiclass voice spoofing classification problems. The use of raw wave front-ends has also been investigated in speaker spoofing detection (Dinkel et al., 2018).
- **Gated Recurrent Units (GRUs):** GRUs are a type of Recurrent Neural network (RNN) that have proven to be useful for audio classification problems, for example, GRUs were found to be suitable for audio scene classification (Phan et al., 2017).

- **Long Short Term Memory (LSTMs):** LSTMs are a type of Recurrent Neural Network (RNN) which are successful for audio classification problems, as in Scarpiniti, Communiello, Uncini, and Lee (2021) for classifying construction machines based on their sound. LSTMs have also proven useful in voice classification problems as in Ertam (2019) for recognising gender based on voice. Based on these promising results, we evaluated the use of LSTMs in the voice spoofing classification problem.

For the creation of models, the Tensorflow Python library (Abadi et al., 2016) was used. Tensorflow has been used in other studies such as Kim, Lee, and Nam (2019), where it was used for the creation of CNN models for audio classification problems. Additionally, Tensorflow offers GPU support which can significantly increase the computation speed in training complex deep learning models. For high computing intensity tasks such as training the deep learning models, or tasks which require a significant amount of time such as model evaluation, Northern Ireland High-Performance Computing's Kelvin2 high-performance computing service¹ was used. GPU nodes running NVIDIA Tesla A100 GPUs were used for model training, and Dell PowerEdge R6525 compute nodes were used for CPU computation tasks. These high-performance services were used primarily due to the quantity of data included in the collated dataset, which significantly increased the time and computation power required in each development phase.

3.3. Audio feature extraction

We utilised the *Librosa* (McFee et al., 2015), a Python package for music and audio analysis for audio feature extraction and audio manipulation. *Librosa*² is primarily designed for music, as demonstrated in Raguraman, Mohan, and Vijayan (2019), but it has also been successfully used for speech emotion recognition systems in Babu, Nagaraju, and Vallabhuni (2021). *Librosa* is an appropriate library for one of the goals of this work, which is to investigate spectral feature correlation to the speech audio classification problem because it also provides a large variety of spectral features that can represent components of audio data. The *Librosa* spectral features are used in two main formats. Firstly, for the CNN implementations, Matplotlib (Hunter, 2007) was used alongside *Librosa* to plot the spectral features extracted from the audio files as RGB graph images. Additionally, the *Librosa* feature data was extracted as Numpy (Harris et al., 2020) arrays for use in the GRU and LSTM models. By using the variety of features as shown in Fig. 2, many aspects of the audio data will be represented by the extracted feature data. The definition of each feature is as follows³:

- **Chroma Energy Normalised Chromagram (CENS):** Represents energy distribution of pitch across 12 pitch classes.
- **Short Term Fourier Transform Chromagram (STFT):** Chroma data extracted from an STFT spectrogram.
- **Constant Q Transform Chromagram (CQT):** Chroma data based on Constant Q time frequencies.
- **Mel Spectrogram:** Spectrogram of audio data presented on the Mel scale.
- **Spectrogram:** Representation of the frequency of an audio signal over time.
- **Mel Frequency Cepstral Coefficients (MFCC):** Representation of the shape of a spectrum based on the Mel Spectrogram.
- **Root Mean Square (RMS):** Data for the energy amplitude of an audio signal.
- **Spectral Bandwidth:** Range of audio signal frequency.

¹ <https://www.qub.ac.uk/directorates/InformationServices/Services/HighPerformanceComputing/#kelvin-cluster-1063431-1>.

² (<https://librosa.org/>).

³ <https://librosa.org/doc/latest/feature.html>.

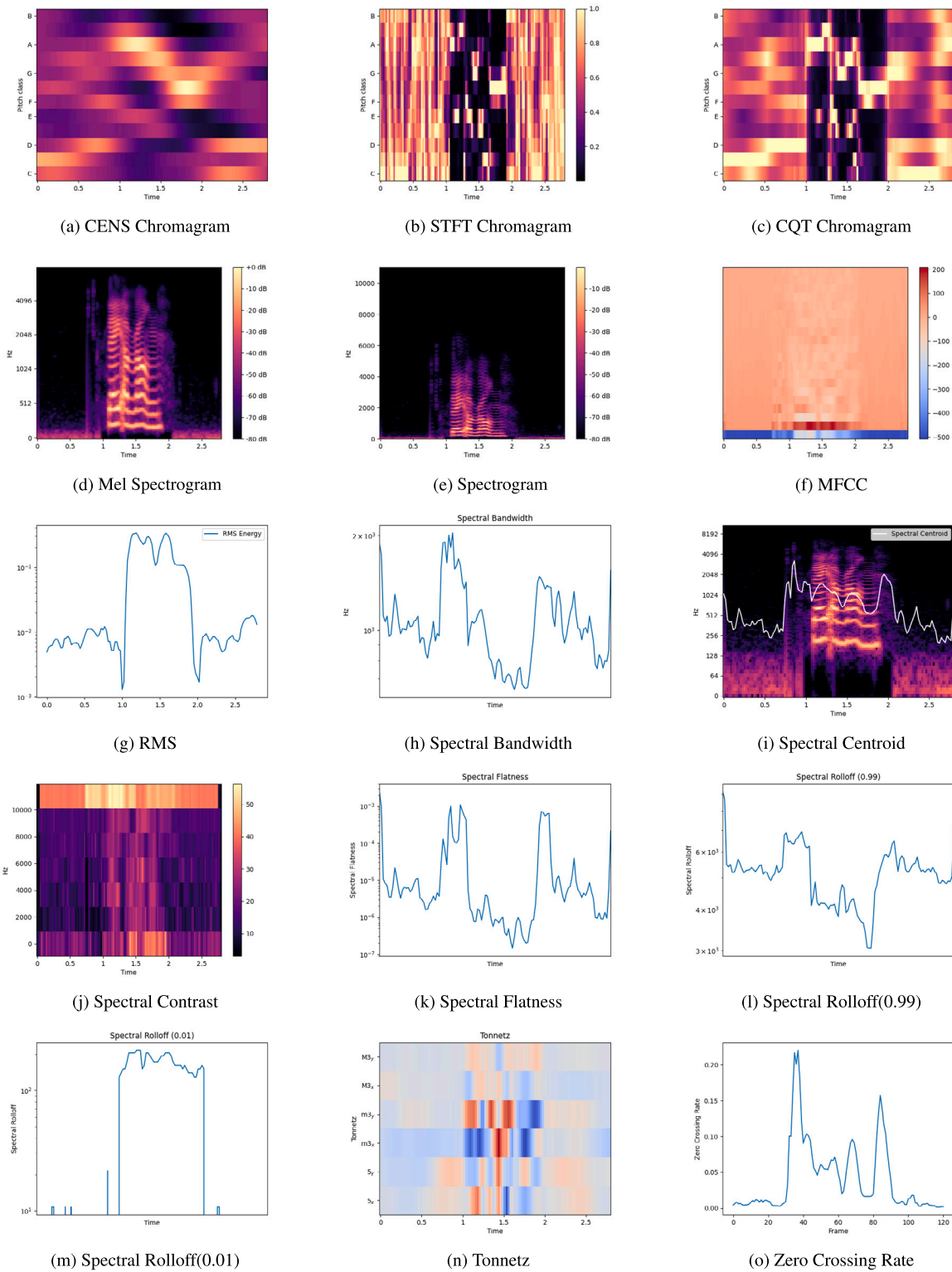


Fig. 2. Librosa spectral features.

- **Spectral Centroid:** Centre of mass of audio frequency along a timescale.
- **Spectral Contrast:** Contrast between amplitudes (peaks and troughs) of the frequency of the audio signal over time.
- **Spectral Flatness:** Measure the flatness of the amplitudes in the audio frequencies.
- **Spectral Rolloff:** Measurement of the frequency spectrum in the audio signal which lays below a certain percentage. In our proposed solution, we utilise both the 1% and 99% for a broader representation of the spectral roll-off.
- **Tonnetz:** Geometric-based representation of pitch.
- **Zero Crossing Rate:** Rate of the audio signal at which it crosses the zero-amplitude value.

Table 4
Sampling rate, duration and file format of dataset.

Dataset	Frequency	Duration	File format
VSDC	96 000 Hz	~6 s	.wav
ASVSpooof2019	16 000 Hz	~2.5 s~5 s	.flac
Deep4SNet	44 100 Hz	~10 s	.wav
ASVSpooof2017	16 000 Hz	~2.5~5 s	.wav
ASVSpooof2015	16 000 Hz	~2.5 s~5 s	.wav

3.4. Normalisation of audio data

The deep learning model algorithms investigated (cf Section 3.2) are trained to expect specific input shapes. An input shape is the dimensions of the input data and can be based on several factors depending on the algorithm. In CNNs, the input shape is based on the resolution of the image inputs, as well as if the images are RGB (red, green, blue colour) or black and white. In this case, the CNN input shape is (255, 255, 3). For WaveNet, the input shape is based on the frequency and duration of the audio files, where the input shape is the frequency multiplied by the duration. In this case, the WaveNet input shape is (48 000,) for 16 kHz audio files of 3 s duration. In the LSTM and GRU models, the input shape is based on the dimensions of the NumPy array containing the extracted feature data. This varies for every feature, but as an example, the MFCC feature has a shape of (20, 94) for a 16 kHz audio file of 3 s duration.

To conform with the expected input shapes for models trained on audio files (WaveNet, GRU, LSTM), the audio data files within the collated binary and multiclass datasets were modified to normalise their duration and frequency to ensure a uniform input shape containing the audio data could be passed to the models. The collated dataset in Table 3 largely consists of the ASVSpooof2015, ASVSpooof2017 and ASVSpooof2019 audio files, which ranged from approximately 3 to 5 s in duration with a frequency of 16 kHz. On the other hand, the Deep4SNet and VSDC audio files were of higher frequencies (44.1 kHz and 96 kHz respectively) and longer in duration (~6 to ~10 s). Most files were in .wav format, but some were in .flac file format. A breakdown of the specific file details is presented in Table 4.

To ensure that the input shapes were consistent across the entire collated dataset, all files were normalised by downsampling their frequency to 16 kHz and reducing their duration to 3 s. In addition, to ensure the file format was consistent across the dataset, the .flac files present in the dataset were converted to .wav format. To normalise the duration, the first 3 s of the audio data were selected if possible. For audio files under 3 s in duration, a padding of constant 0 was applied to the end of the audio files until the duration became 3 s. The process of downsampling was performed using Librosa by first loading the audio and original sampling rate, then using the `Librosa.resample()` method to rewrite the audio with 16 kHz frequency. The newly uniform duration and downsampled audio were then written to a new audio file using FFmpeg (Tomar, 2006). FFmpeg is a command line tool that enables the conversion of file formats, and in this case, was used to convert the .flac files of the ASVSpooof2019 dataset to .wav files.

3.5. Performance metrics

To evaluate the CNN, WaveNet, GRU and LSTM models several performance metrics have been considered. Firstly, the classification report of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) will be recorded to see where the classification is failing or succeeding. To further supplement this, the **Precision** (ratio of true positives out of all positives), **Recall** (ratio of true positives out of all actual positives), **False positive Rate(FPR)** (percentage of false positives out of all positive predictions), and **F1 score** (harmonic mean of precision and recall) will be reported. Additional performance measures that account for class imbalance such as **Matthew's Correlation Coefficient (MCC)**, **Cohen's Kappa Score (CKS)**, and **Balanced**

Accuracy (BA) are also used to assess the classifiers' performance. MCC measures the correlation between the predicted class and the actual class based on the TP, FP, TN, and FN. CKS is an evaluation metric that calculates the agreement between the predicted class and the true class while making considerations of the potential agreement of predicted and actual labels by chance. Balanced accuracy is a sensitivity-specificity metric obtained by calculating the arithmetic means of sensitivity (True Positive Rate) and specificity (True Negative Rate). These performance measures are benchmark recommendations for avoiding base rate fallacy in detection tasks (Arp et al., 2022; Branco, Torgo, & Ribeiro, 2016). The performance measures for evaluating the deep learning classifiers were computed using the *Scikit-learn* python library (Pedregosa et al., 2011). In several studies, including Ali, Zhu, Zhou, and Liu (2019), which uses voice recordings for diagnosing Parkinson's disease, and in Dhakal, Damacharla, Javaid, and Devabhaktuni (2019), which employs deep learning for automatic speaker recognition, *Scikit-learn* has been demonstrated to be useful in audio analysis tasks.

4. Evaluation

In this research, the voice spoofing detection problem was investigated as a binary and a multiclass problem using deep learning algorithms. Binary classification models capable of classifying spoofed audio against bonafide recordings were developed. With the multiclass models, the classes of attack are detected against bonafide audio recordings. To evaluate the models, different types of audio data representing different classes of attacks and diverse conditions were collated. The models were evaluated using a dataset of pre-recorded audio files collated from benchmark datasets in literature for voice spoofing detection (see Section 3.1), novel audio data from the microphone and user-supplied inputs via the web services of the deployed model. To verify the contributions of the proposed research, here are the questions, the evaluation of the proposed framework aims to answer.

- **RQ1 — Feature Evaluation:** What are the most crucial spectral features needed for binary and multiclass voice spoofing detection?
- **RQ2 — Diverse Algorithms:** What insights can we draw from investigating the effectiveness of multiple deep learning algorithms against voice spoofing attacks to that of a single algorithm?
- **RQ3 — Diverse Inputs** What impact do the three separate inputs – pre-recorded audio files, original audio captured by a microphone and user input – have on the proposed framework?
- **RQ4 — Comparison With State-of-the-Art** How does the proposed approach contribute to knowledge in voice spoofing countermeasures using deep learning?

4.1. RQ1 - Feature evaluation

To answer RQ1, we evaluated spectral features and identified which features are crucial to the binary and multiclass classification task in voice spoofing detection using convoluted neural networks. Generally, CNNs are image-based classifiers, so the audio files from the training dataset were extracted as feature graphs and exported as RGB .png files as shown in Fig. 2 (see Section 3.3). These PNG images were then fed to the classifier, allowing it to learn patterns in the feature images for prediction. As described in Section 3, a train test ratio of 70/30 was used for the training data, using the Tensorflow Python library to define and train the models. The sequential CNN model in both binary and multiclass classification is shown in Fig. 3.

To investigate the feature correlation of each of the features, a CNN model was fitted for every feature. This allowed the significance of each feature to be determined based on the performance of the model. The performance of each feature is reported for the binary classification in Table 5, and the multiclass classification in Table 6. Additionally, to

Table 5
Feature evaluation using CNN for binary classification of spoofed vs. bonafide audio.

Feature	FPR	Precision	Recall	F1 score	BA	MCC	CKS
Spectral Centroid	0.0231	0.9876	0.9875	0.9875	0.9730	0.9430	0.9429
MelSpectrogram	0.0645	0.9763	0.9766	0.9763	0.9313	0.8903	0.8896
Spectrogram	0.0932	0.9530	0.9529	0.9530	0.8931	0.7846	0.7845
Chromagram	0.0979	0.9192	0.8970	0.9045	0.8572	0.6157	0.6020
RMS	0.1698	0.9252	0.9280	0.9263	0.8132	0.6560	0.6548
Spectral Contrast	0.2134	0.9399	0.9408	0.9347	0.7831	0.6998	0.6761
Spectral Flatness	0.2220	0.9264	0.9311	0.9251	0.7697	0.6420	0.6270
ZCR	0.2157	0.9071	0.9119	0.9090	0.7642	0.5652	0.5631
CQTChromagram	0.2534	0.8961	0.9043	0.8988	0.7274	0.5108	0.5053
Spectral Bandwidth	0.2695	0.8405	0.7933	0.8129	0.6503	0.2477	0.2381
Spectral Rolloff 99%	0.3349	0.8530	0.8676	0.8590	0.6364	0.3112	0.3069
Tonnetz	0.3456	0.8553	0.8738	0.8619	0.6308	0.3170	0.3081
CENSChromagram	0.3987	0.8744	0.8917	0.8645	0.5952	0.3375	0.2743
Spectral Rolloff 1%	0.4739	0.8761	0.8826	0.8345	0.5253	0.1895	0.0848
MFCC	0.4836	0.8838	0.8791	0.8268	0.5162	0.1605	0.0551

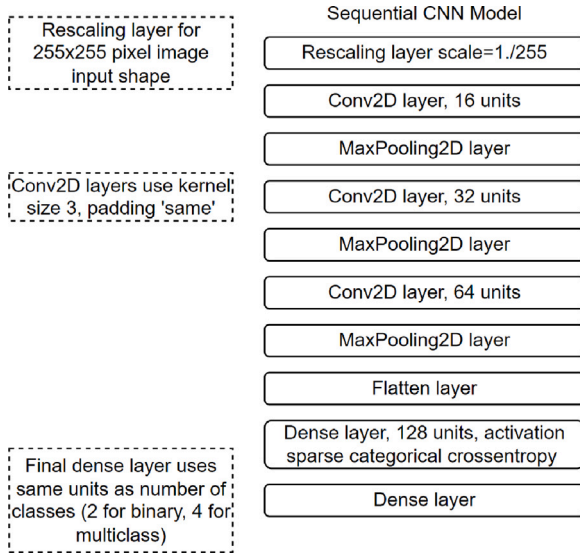


Fig. 3. The architecture of the CNN model.

determine if the CNN classifier is suitable for both binary and multiclass classification where 'binary' is the classification of audio as bonafide or spoofed (spoofed class is a combination of converted, replayed, and synthesised audio) and multiclass is the classification of audio as real, converted, synthesised, or replayed, both binary and multi-class models were trained. The purpose of the binary classifier is simply to identify if audio is real or not, while the purpose of the multi-class classifier is to identify if the audio is real or, if not, which of the attack vectors (conversion, replay, synthesis) is employed in the audio.

In our analysis, we performed the experiments over features extracted from the audio data that ranges from CENS chromagram to zero crossing rate. However, two or more features may combine to create a new feature that may be significant, with the combination being additive or multiplicative. The expectation is that the combination of multiple features can provide more information to efficiently learn the simple model (i.e., CNN). The hypothesis was to extract the possible feature and combination of features which will aid the model in distinguishing between real and spoofed audio (binary classification), as well as the sophistication of the attack (multiclass classification). The Spectral Centroid and MelSpectrogram, with false positive rates of 0.0231 and 0.0645, respectively, were the best-performing features in the binary classification of faked audio against authentic speech recordings, as shown in Table 5. An additional model was developed and trained to utilise both features to test whether their combination would improve performance. The F1 score for the combination of the

Spectral Centroid and MelSpectrogram model was 0.9844, the Balanced Accuracy was 0.9732, the MCC was 0.9756, and the CKS was 0.9754. These outcomes show that the combination outperformed the classifier trained on the MelSpectrogram alone in every way. However, the combination of these features only improved the MCC and CKS; it did not, however, improve the F1-Score or the balanced accuracy of the classifier trained on the Spectral Centroid alone. As the results are not statistically significant, this finding suggests that the Spectral Centroid is the best feature for binary classification using CNN to produce a near-perfect classification of spoofed audio against bonafide voice recordings.

The top three performing features in the classification of replay, synthesised, and converted attacks against real audio were the Spectral Centroid, MelSpectrogram, and Spectral Contrast, as shown in the multiclass classification performance findings in Table 6. An additional multiclass model was fitted and trained on both the Spectral Centroid and MelSpectrogram due to the performance of the Spectral Centroid and MelSpectrogram being quite close. The performance results of the multiclass model trained on the combination of the features were an F1 score of 0.9833, balanced accuracy of 0.9681, MCC of 0.9772, and CKS of 0.9774. The multiclass model trained on Spectral Centroid and MelSpectrogram performance outperformed the model trained on MelSpectrogram alone, as was seen in the binary classification problem. However, the performance of the model trained on the Spectral Centroid alone was better than the performance of the feature combination model in terms of F1-Score (0.9864 vs. 0.9833), Balanced Accuracy (0.9791 vs. 0.9681), MCC (0.9797 vs. 0.9772), and CKS (0.9797 vs. 0.9774). The results suggested that the Spectral Centroid is the best-performing feature for both the binary and multiclass classification problem in voice spoofing detection, obtaining a near-perfect classification across the testing set.

Hyperparameter tuning for binary and multiclass classification

Hyperparameter adjustment was performed to make the binary and multiclass Spectral Centroid CNN models as effective as possible. Experimentation targeted layer configuration through increasing or decreasing units and kernel size, as well as increasing and decreasing the batch size. It was found that a model with double-layer filter units compared to the default model performed best. Batch size and kernel size had no significant impact on the classification ability, so the default settings for these were retained. During training, the callbacks at each epoch were monitored to prevent overfitting to the training data, and after 12 epochs the training came to an early stop.

As shown in Table 7, the optimal spectral centroid CNN for the binary classification of spoofed audio against bonafide recordings has an almost near-perfect prediction of classes reinforced by an F1-Score of 0.9916. With an FPR of 0.0216, this shows the ability of the model to classify real and fake audio, even with the large class imbalance

Table 6

Feature evaluation using CNN for multiclass classification of replay, converted & synthesised vs. bonafide audio.

Feature	FPR	Precision	Recall	F1 score	BA	MCC	CKS
Spectral Centroid	0.0046	0.9864	0.9864	0.9864	0.9791	0.9797	0.9797
MelSpectrogram	0.0080	0.9804	0.9805	0.9801	0.9659	0.9707	0.9706
Spectral Contrast	0.0162	0.9550	0.9506	0.9512	0.9256	0.9267	0.9260
Spectrogram	0.0251	0.9412	0.9412	0.9388	0.8901	0.9136	0.9114
CQTChromagram	0.0270	0.9256	0.9247	0.9251	0.8907	0.8861	0.8861
Chromagram	0.0361	0.8968	0.8874	0.8910	0.8495	0.8339	0.8332
ZCR	0.0357	0.8917	0.8908	0.8908	0.8424	0.8364	0.8360
RMS	0.0388	0.8888	0.8967	0.8862	0.8194	0.8453	0.8424
MFCC	0.0425	0.8759	0.8883	0.8773	0.8117	0.8321	0.8296
CENSChromagram	0.0528	0.8621	0.8725	0.8561	0.7760	0.8066	0.8005
Spectral Bandwidth	0.0594	0.8429	0.8351	0.8359	0.7613	0.7508	0.7494
Tonnetz	0.0581	0.8413	0.8525	0.8441	0.7582	0.7736	0.7710
Spectral Rolloff 99%	0.0641	0.8323	0.8133	0.8179	0.7479	0.7226	0.7200
Spectral Flatness	0.0648	0.8339	0.8019	0.8111	0.7324	0.7076	0.7039
Spectral Rolloff 1%	0.1864	0.7276	0.6196	0.5566	0.4223	0.4076	0.3037

Table 7

Hyperparameter settings and performance results for optimal spectral centroid CNN.

Hyperparameter settings		Optimal performance results	
Parameter	Values	Binary classification	Multiclass classification
Batch size	32	FPR = 0.0216	FPR = 0.003
Optimiser	Adam	MCC = 0.9614	MCC = 0.9897
Epochs	12	BA = 0.9767	BA = 0.9857
Trainable params	8,412,836	F1 = 0.9916	F1 = 0.9919

Table 8

Hyperparameter settings and performance results for multiclass WaveNet.

Hyperparameter for optimal WaveNet model		Optimal performance results
Parameter	Values	Multiclass classification
Batch size	8	FPR = 0.0260
Optimiser	Adam	MCC = 0.9070
Epochs	80	BA = 0.8987
Kernel size	3	F1 = 0.9320
Dilation depth	3	
Number of filters	32	

between the real and spoofed dataset, with bonafide audio making only approximately 12% of the entire dataset, with the remaining 88% spoofed (see Table 3). Also, the tuned optimal binary spectral centroid CNN (cf Table 7) yielded a lower FPR compared with the default parameters used in Table 5. For the multiclass classification problem, Table 7 also shows that the optimal Spectral Centroid has improved performance with an FPR of 0.3% compared to 0.46% with default parameter settings in Table 6. The F1 score improved throughout the classification in each class with F1 scores of 0.9875 for converted, 0.9682 for real, 0.9955 for replay, and 0.9993 for synthesised audio. Additionally, the false positive rate of 0.003 is very low, and although there is a class imbalance in the dataset with 57% of the spoofed audio in the replay attack class (see Table 3). The imbalance impartial metrics also indicate that the model performs with a balanced accuracy of 0.9857, and MCC 0.9879.

This experiment into the suitability of CNNs in the binary and multiclass problem in voice spoofing detection suggests that CNN models are useful in detecting voice spoofing. The second finding is that Spectral Centroid is a crucial feature in detecting spoofed audio against bonafide audio; and for further investigation into the type of attack in the spoofed audio.

4.2. RQ2 - Diversity of algorithms

To evaluate the impact of using diverse deep learning algorithms on voice spoofing detection, we investigate the suitability of WaveNet, LSTM and GRU on the multiclass classification problem.

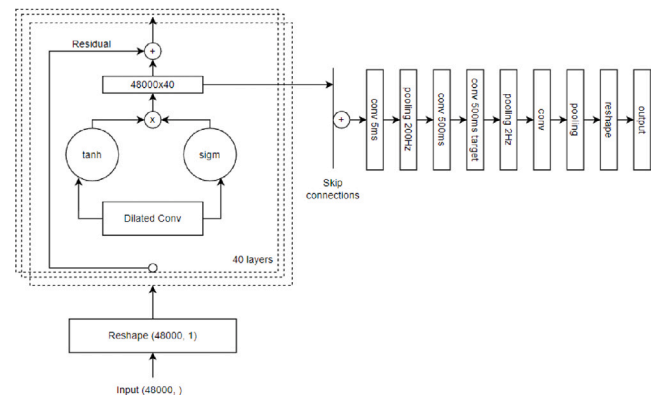


Fig. 4. The architecture of the WaveNet based deep model (Zhang et al., 2020).

WaveNet

As discussed in Section 3.2, WaveNet (Oord et al., 2016) is a deep generative model of raw audio waveforms that are trained on speech samples.⁴ For the multiclass classification problem in voice spoofing detection, a WaveNet model was implemented. Since the WaveNet is trained on raw waveform audio signals, feature extraction was not necessary for this instance. The audio files from the dataset were, however, normalised to have a consistent input shape, with frequency resampled to 16 kHz, and audio files clipped to 3 s in length, as indicated in Section 3.4.

Fig. 4 shows the architecture of the WaveNet classification model as in Zhang et al. (2020), implemented using Tensorflow, and using the same 70/30 split for the train/test data. The dilated convolutional layers and gated activation units are beneficial for increasing the ability of the model to learn complex sequences in the audio data. Experimentation of hyperparameter settings was carried out to investigate how different hyperparameters may affect the classification performance of the multiclass models. Batch size, kernel size, dilation depth, and number of filters were investigated. Based on the train/test accuracies during model fitting, increased dilation depth and kernel size did not affect accuracy compared to the initial model. However, it was found that a batch size of 8 with 32 filters was optimal and increased accuracy by approximately 5%, so it was concluded that the optimal hyperparameters were found. The hyperparameters for the optimal model are displayed in Table 8.

Using these optimal hyperparameters, the multi-class WaveNet models were fitted achieving performance metric values such as FPR of

⁴ <https://www.deepmind.com/research/highlighted-research/wavenet>.

Table 9

Feature evaluation based on performance of multiclass GRU with optimal hyperparameters.

Feature	FPR	Precision	Recall	F1 score	BA	MCC	CKS
Spectral Contrast	0.0408	0.8777	0.8823	0.8754	0.8060	0.8231	0.8165
MFCC	0.0485	0.8712	0.8815	0.8728	0.7924	0.8181	0.8149
STFT Chromagram	0.0552	0.8402	0.8588	0.8439	0.7636	0.7841	0.7808
CQT Chromagram	0.0614	0.8254	0.8457	0.8238	0.7451	0.7638	0.7594
CENS Chromagram	0.0659	0.8131	0.8336	0.8160	0.7324	0.7450	0.7406
RMS	0.0750	0.7952	0.8187	0.7961	0.7029	0.7206	0.7134
Spectral Centroid	0.0850	0.7713	0.7968	0.7700	0.6703	0.6853	0.6760
Spectral Bandwidth	0.0948	0.7512	0.7770	0.7466	0.6389	0.6527	0.6405
Spectral Flatness	0.0968	0.7467	0.7706	0.7414	0.6324	0.6421	0.6305
Spectral Rolloff	0.1025	0.7438	0.7585	0.7273	0.6170	0.62210	0.6094
Tonnetz	0.1027	0.7339	0.7582	0.7260	0.6165	0.6218	0.6090
ZCR	0.1073	0.7242	0.7482	0.7144	0.6034	0.6051	0.5913

Table 10

Hyperparameter settings and performance results for optimal multiclass GRU.

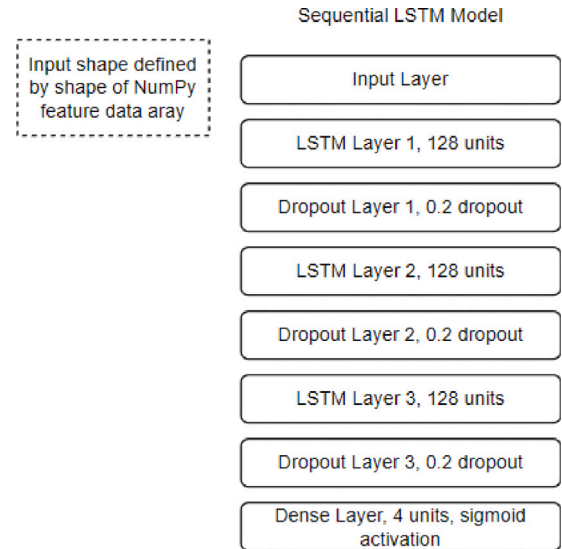
Hyperparameter for optimal GRU model		Optimal performance results
Parameter	Values	Multiclass classification
Batch size	8	FPR = 0.0358
Optimiser	Adam	MCC = 0.8549
Epochs	34 (early stop)	BA = 0.8476
Trainable params	284,676	F1 = 0.9015
Learning rate	0.000001	

0.026 and a balanced accuracy of 0.9070. However, it was observed that the model is best at classifying synthesised, converted, and replay audio (F1 scores in the range of 0.9399 to 0.997) but less capable of predicting bonafide audio (F1 score of 0.7212). Nonetheless, each class is still predicted at rates greater than chance, with an average F1-score of 0.9320, and it can be concluded that WaveNet is a suitable algorithm for the multiclass problem in voice spoofing detection

Gated Recurrent Units (GRU)

GRU models were implemented using the spectral features described in Section 3.3. Similarly to CNNs and WaveNet, this experiment also uses the Tensorflow Python library to define and train the GRU models, with a 70/30 train/test split of the collated data. To investigate feature correlation to the classification problems, multiclass models were fitted on single audio features. In this case, the features were extracted as NumPy arrays of float values and the models were trained on those. Hyperparameter tuning was performed to investigate how different model setups may benefit training and testing accuracies. We experimented with batch size, layer architecture, and the number of filters per GRU layer. A batch size of 8 with 128 filters for each GRU layer was found to boost train/test accuracy by about 5%. Additionally, the learning rate was decreased and dropout layers were added to the model architecture to reduce overfitting to the training set of data. Table 9 shows the results for the multiclass GRU feature models.

The best-performing features were the MFCC and Spectral Contrast, which both shared very similar evaluative metrics as shown in Table 9. To investigate if a combination of these features would be better than how they perform independently, a new GRU model was fitted with the optimal hyperparameter settings shown in Table 10 using both MFCC and Spectral Contrast as training features. While a balanced accuracy of 0.8476 and MCC of 0.8549 shows the classifiers' performance across an imbalanced dataset, the model is limited by its classification of bonafide audio, where it only achieves a precision of 0.6659, recall of 0.5547, and F1-score of 0.6052 in contrast to the higher precision-recall metrics value in classifying the replay, synthesis and conversion attack as shown in the optimal performance results.

**Fig. 5.** The architecture of the LSTM model.

Long Short-Term Memory Networks (LSTM)

The last algorithm under investigation for voice spoofing detection as a multiclass problem is LSTMs fitted using spectral features. To define the model architecture and train the models, Tensorflow was used with a 70/30 train test split. The model architecture for the LSTM is displayed in Fig. 5. The results for the multi-class models trained on single audio features are shown in Table 11. These performance metrics indicate that the MFCC and Spectral Contrast features hold a high correlation, even when accounting for the imbalance in the dataset.

To investigate if a combination of these features would produce better results, a new model was fitted using the same model architecture and hyperparameters (cf Table 12) using a combination of MFCC and Spectral Contrast as training features. However, when evaluated it was observed that this model's performance dropped significantly with a FPR of 0.0703, precision of 0.8205, recall of 0.8532, F1 score of 0.8187, balanced accuracy of 0.7201, MCC of 0.7457, and CKS of 0.7380. The performance of the model therefore reduced by approximately 9% to 10% in each of the evaluative metrics, so the combination of features was not successful. In an attempt to improve the classification ability of the MFCC model further, hyperparameter tuning was performed. Batch size, layer units, and layer configuration were investigated. It was found that a batch size of 8, 128 units per layer and the addition of dropout layers to the model resulted in a train/test accuracy increase of around 5%.

The optimal model performance results in Table 12 indicate that the model makes little error in the multiclass classification of voice spoofing attacks across an imbalanced dataset. However, the model

Table 11

Feature evaluation based on performance of multiclass LSTM with optimal hyperparameters.

Feature	FPR	Precision	Recall	F1 score	BA	MCC	CKS
MFCC	0.0420	0.8809	0.8853	0.8819	0.8323	0.8264	0.8259
Spectral Contrast	0.0432	0.8717	0.8699	0.8679	0.8229	0.8075	0.8056
CQT Chromagram	0.0591	0.8369	0.8570	0.8398	0.7533	0.7807	0.7753
STFT Chromagram	0.0576	0.8285	0.8535	0.8255	0.7482	0.7792	0.7718
Tonnetz	0.0735	0.7893	0.8185	0.7958	0.7026	0.7205	0.7146
CENS Chromagram	0.0734	0.7354	0.8316	0.7793	0.6953	0.7469	0.7298
RMS	0.1006	0.7694	0.7733	0.7263	0.6234	0.6490	0.6294
Spectral Centroid	0.2023	0.5452	0.5805	0.4939	0.3587	0.3002	0.2226
Spectral Bandwidth	0.2002	0.4175	0.5688	0.4722	0.3467	0.2657	0.2202
Spectral Rolloff	0.2152	0.4812	0.5257	0.4522	0.3329	0.1824	0.1588
Spectral Flatness	0.2500	0.2491	0.4991	0.3323	0.2500	0.0000	0.0000
ZCR	0.2500	0.2491	0.4991	0.3323	0.2500	0.0000	0.0000

Table 12

Hyperparameter settings and performance results for multiclass LSTM.

Hyperparameters for optimal LSTM MFCC model		Optimal performance results
Parameter	Values	Multiclass classification
Batch size	8	FPR = 0.0302
Optimiser	Adam	MCC = 0.8698
Epochs	24 (early stop)	BA = 0.8934
Trainable params	140,420	F1 = 0.9148

performs better at classifying synthesis, conversion and replay attacks, and worse at classifying real audio. In classifying bonafide audio from spoofed audio with replay, conversion and synthesis attacks, the model achieves this with a precision of 0.6430, recall of 0.7276, and F1 score of 0.6827. However, the model does perform better than chance in the classification of bonafide audio.

With the investigation of the four deep learning algorithms, we can conclude that CNN, WaveNet, GRU and LSTM are suitable algorithms for the multiclass problem in voice spoofing detection using the optimal performance metrics (cf Tables 7, 8, 10 and 12). To draw insights from investigating the effectiveness of these multiple deep learning algorithms against voice spoofing attacks to that of a single algorithm using their optimal features, we investigate the performance of the precision-recall metrics of each across each class of real, synthesised, converted and replays as a multi-class problem. Based on the results of the precision-recall metrics in Table 13, the overall performance values of CNN and WaveNet exceed every performance metric of GRU and LSTM. While the CNN metrics showed a better overall performance than the WaveNet model, the precision-recall metric of the synthesised class is higher than that of the CNN Model, which suggests that the WaveNet algorithm is better at detecting synthesised attacks. The implication of this is that the CNN and the WaveNet model can be combined in ensemble learning to achieve an optimal performance since they use different forms of input. However, the performance of the WaveNet model in classifying real audio is considerably lower compared with the CNN model, as it only correctly identified 69% of the real audio files as real, confusing 26% for replay and 4% for converted audio. Similarly to WaveNet, the LSTMs do struggle with the classification of real audio, which only correctly predicted 73% of the real audio files, confusing 20% for replay audio, and 7% for converted audio. Amongst the four algorithms investigated, GRUs were the least of the algorithms investigated but still had acceptable performance across the evaluation set. The GRU model is similar in performance to the LSTM model albeit slightly worse overall, which could indicate that recurrent neural networks, which GRUs and LSTMs are variants of, may be less suitable for multiclass classification than other deep learning algorithms like CNNs and WaveNet. The optimal GRU model only predicted 55% of the real audio data correctly and predicted 30% as a replay, 14% as converted, and 1% as synthesised. This shows that the classification of real audio is only slightly better than chance, and the GRU model is unreliable for the classification of real audio. In summary, to answer the research

Table 13

Performance comparison and rank of CNN, WaveNet, GRU & LSTM in multiclass classification.

Model	Metrics	Replay	Synthesised	Converted	Real	Rank
CNN	Precision	0.9836	0.9830	0.9928	0.9996	1
	Recall	0.9915	0.9539	0.9982	0.9990	
WaveNet	Precision	0.9360	0.9966	0.9560	0.7556	2
	Recall	0.9440	0.9976	0.9968	0.6898	
GRU	Precision	0.9235	0.9881	0.8693	0.6659	4
	Recall	0.9524	0.9933	0.8902	0.5547	
LSTM	Precision	0.9486	0.9925	0.9198	0.6430	3
	Recall	0.9108	0.9969	0.9383	0.7276	

question into the evaluation of a variety of deep learning algorithms for detecting voice spoofing, it can be concluded that CNNs, WaveNet and LSTMs are suitable for voice spoofing detection. However, the inability of GRUs to reliably classify real audio makes them less suitable for voice spoofing countermeasures.

This research domain is important for automated speaker verification or voice biometric authentication systems as malicious actors employ diverse types of attacks such as speech synthesis, voice conversion or imitations, and recorded replays to spoof these authentication systems. The detailed empirical analysis of the designed deep learning models confirms the advantages and preferences over one another in the domain of audio spoofing. The CNN model is light as compared with WaveNet and LSTM/GRUs. Such model has the advantage of being deployed in resource constrained devices. It achieves better performance for binary and multi-class learning. The WaveNet learning model processes the raw audio data which eliminates the need for feature extraction required for CNN, LSTM & GRU. The goal was to check the extracted features by mapping temporal dependencies with GRU & LSTM for multiclass learning. These models are well-known to capture the long dependencies which are aligned to audio data. In particular, the temporal dependency in audio data is an innate characteristic that has already been widely adopted in the machine learning models and has also been exploited to characterise audio adversarial samples (Yang, Li, Chen, & Song, 2018). All the models perform well, and the choice can be made over the existing hardware. For instance, CNN proves to be the most preferable model over the embedded devices used in ML-powered voice analysis to provide real-time caller authentication and fraud risk detection. In case of WaveNet, LSTMs and GRUs, it computes more parameters for making inferences, which results in high utilisation of CPU/GPU resources. Such kind of models are used in multiple experts setting, where an individual audio requires confirmation from more than one AI model to increase confidence of the classified audio data. These multiple expert settings are finding increasing adoption in interactive machine learning through jury learning (Gordon et al., 2022) and majority voting ensemble approach (Atallah & Al-Mousa, 2019; Dogan & Birant, 2019), where multiple machine learning models are combined to determine the label and/or prediction result in used in classification tasks.

4.3. RQ3 - Diverse inputs

In RQ1 (cf Section 4.1) and RQ2 (cf Section 3.2), we evaluate the deep learning algorithms on pre-recorded voice recordings from the collated Dataset in Table 3. To further evaluate the models, we subject the testing to novel and unseen audio capture by the microphone for real-time multiclass classification. We also deployed the models as a web application for evaluation using user input.

Real-time classification using a microphone

To evaluate the performance of the models to the multiclass classification problem in voice spoofing detection in real-time, experimentation was conducted to record voice input directly from a microphone, process the recorded audio, and then pass the processed audio to the optimal CNN, WaveNet, GRU, and LSTM models for classification. As the real-time audio was unique and not included in the training or evaluation dataset, this was an interesting test to assess how the models perform on completely new and unknown data. Additionally, this experiment is based on a potential real-life application of the created models, for example in automated speaker verification systems or biometric authentication, so offered an insight into how the models may perform if deployed in software systems.

To capture the audio in real-time, the PyAudio library,⁵ was used and enabled recording the data from the audio stream of the microphone into a suitable format, frequency, and duration. The capture takes place over 3 s, at a frequency of 16 kHz so that the input shape is valid for the WaveNet model. Data processing is done to firstly pad the captured audio so that it is exactly 3 s in duration, then to extract the relevant features from the audio for use in the models. The same padding and feature extraction techniques are employed for real-time processing as were used in the subsequent sections for training the models (cf Sections 3.3, 3.4). After capturing the raw audio and extracting features, the audio is passed to each of the models which returns a prediction for the label of the audio.

In terms of hardware, the audio was captured using a Blue Yeti microphone and replayed from the speakers of an OPPO Find X2 Lite mobile phone. For conversion attacks, Logitech G Hub was used to apply several Blue 'VO!CE EFFECTS'⁶ to the microphone capture in real-time. The choice of hardware used for the system architecture in the real-time classification of voice spoofing attacks is based on several reviews⁷ that put the Blue Yeti microphone and Logitech G hub as some of the best microphones for recording audio. As synthesised audio is generated by the software and not produced through a microphone, it was not possible to test it as real-time audio input. However, it was felt that it was still important to test the models on novel synthesised audio data (not from a dataset the model was trained on), so new synthesised audio files were generated using Google text-to-speech and fed to the models for classification. 20 phrases were tested in total, of which 15 were real and 5 spoofed in multiclass classification.

Table 14 shows the real-time multiclass classification performance of the real-time audio by each model. The CNN models performed best with 15/20 accurate predictions. However, of the 6 spoofed classes, only 2 (one synthesis and one replay) were correctly classified. With 12/20 correct predictions, WaveNet was ranked second. Of the 6 spoofed classes, only 3 were correctly classified (one replay, two synthesised). Similar to the performance on the testing dataset (cf Table 13), where the WaveNet performance metrics (Recall) were

better than the CNN on the converted attack class, WaveNet also predicted some attack class better than the CNN model. With 7/20 accurate predictions and 4 of the spoofed classes correctly classified (two replays and two synthesised), LSTMs were ranked third. The least accurate predictions were made by GRUs (6/20), and just 4 of the spoofed classes (two replays, two synthesised) were correctly classified.

The effectiveness of each model is consistent with the findings on the evaluation dataset (cf Table 13), where CNNs performed best, followed by WaveNet, LSTMs and GRUs, a ranking consistent with the results obtained on the testing set. Furthermore, in the real-time classification of microphone input, CNNs accurately predicted 13 out of the 15 real classes, with WaveNet ranking second with only 9 out of 15. This is also consistent with the findings from the testing dataset, where the CNN model outperformed other algorithms in terms of recall, precision, and F1-score when used on real audio, with WaveNet ranking second (cf Table 13). One significant insight from the evaluation on inputs from the microphone is that the patterns observed in the pre-recorded collated dataset were replicated in the real-time classification results, which shows the validation of the results obtained from the dataset evaluation.

However, the accuracy is not consistent with how the models performed on the evaluation dataset, for example, CNNs had near-perfect prediction on the evaluation data, but only successfully predicted 75% correctly in real-time. It is difficult to determine if this is a limitation of the models on unseen audio data, or if it is due to the hardware and software used to capture, convert, and replay the audio. Security research has observed similar discrepancies between classifier performance in the lab and in the wild for some time and this was attributed to sampling bias, which occurs when the collected data does not sufficiently represent the true data distribution of the underlying security problem (Arp et al., 2022). Additionally, there was only a single speaker across the 20 phrases which may have led to decreased performance compared to the evaluation data which was created using multiple speakers and multiple software systems and hardware devices (cf Table 2). Also, the scope of the audio used in the real-time classification experiment is comparatively much smaller than the collated dataset the models were trained on. For example, the real-time classification experiment only made use of a single input and output device and only explored one type of conversion software whereas the collated dataset contained audio recorded and outputted using a wide variety of hardware devices, in multiple settings, and with a range of software conversion and synthesis approaches. Additional experimentation with hardware, software and physical settings may produce better results for real-time classification in real use cases such as automated speaker verification systems.

One area of future work is investigating the effect of speakers, recording devices, recording microphones, playback devices, synthesis algorithms, environments and other audio-capturing devices on voice spoofing detection. The performance of models in the real-time classification of audio captured on the microphone can be affected by the hardware, software, and speakers used to capture, convert, and replay the audio or limitations of the models. This area of research is worthy of investigation for building robust models.

Deployment of the model

To allow users to check if audio is spoofed or real, the optimal models created in the previous sections were deployed to a website using the methodology portrayed in Fig. 6. In the front end, users can interact with the site and select an audio file from their computer. The user can then submit the audio file which creates a HTTP request and sends it to a Python backend. The audio file submission interface is displayed in Fig. 7. The Python backend application was created using Flask to handle the HTTP requests. Upon receiving a request, the application normalises the audio file, extracts the relevant features, and then feeds the features and raw audio into the classification models.

⁵ <https://pypi.org/project/PyAudio/>.

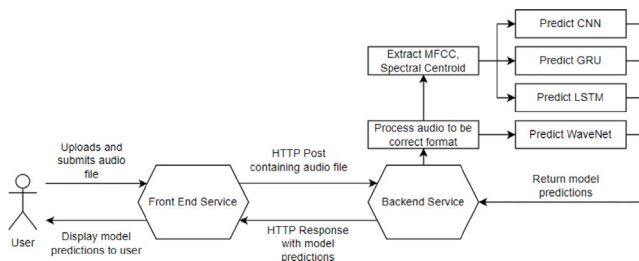
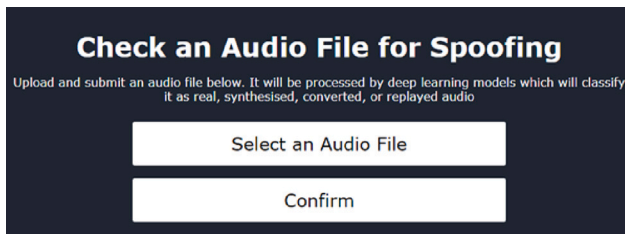
⁶ <https://www.logitech.com/en-gb/innovation/blue-voice-technology.html>.

⁷ <https://computerstationnation.com/best-headset-microphone-for-recording-audio/>, <https://computerstationnation.com/microphones/>, <https://computerstationnation.com/best-usb-microphone-for-streaming/>.

Table 14

Performance of multiclass classification using CNN, WaveNet, GRU and LSTM on microphone inputs.

SN	Phrase	Actual	CNN	WaveNet	GRU	LSTM
1	"Hey Google, unlock the front door."	Converted	Replay	Replay	Replay	Replay
2	"Alexa, disable the security system."	Real	Real	Real	Replay	Real
3	"Hey Google, open the garage door."	Replay	Real	Replay	Replay	Replay
4	"Hey Google, send \$1000 to John using Google Pay."	Real	Real	Replay	Replay	Replay
5	"Alexa, turn off two-factor authentication for my Amazon account."	Real	Real	Real	Replay	Synthesised
6	"Combatting voice spoofing."	Real	Real	Replay	Synthesised	Replay
7	"Binary classification of voice audio."	Real	Real	Real	Real	Real
8	"Jason Boyd."	Synthesis	Replay	Synthesised	Synthesised	Synthesised
9	"Queen's University of Belfast."	Real	Real	Real	Replay	Replay
10	"Multi-class attack vector classification."	Real	Real	Real	Replay	Replay
11	"Replay, synthesis, conversion."	Real	Replay	Replay	Replay	Replay
12	"That's not part of the deal."	Replay	Replay	Real	Replay	Replay
13	"We have the location which is assured."	Converted	Replay	Replay	Replay	Replay
14	"Something is wrong."	Real	Real	Replay	Replay	Real
15	"What will happen then?"	Real	Real	Real	Real	Synthesised
16	"Actions speak louder than words."	Real	Real	Real	Replay	Replay
17	"Necessity is the mother of invention."	Real	Real	Replay	Replay	Replay
18	"My first job was in a hardware store."	Synthesis	Synthesised	Synthesised	Synthesised	Synthesised
19	"Listen to this."	Real	Real	Real	Replay	Replay
20	"It was highly successful."	Real	Real	Real	Replay	Replay

**Fig. 6.** Model deployment architecture.**Fig. 8.** Classification report of a submitted audio file containing converted voice.**Fig. 7.** Front end user interface for uploading the audio file.

This procedure is the same as the methodology developed in the real-time classification using capturing audio input from the microphone. The classification report is then returned by aggregating the prediction of each model and then returned as the HTTP response to be displayed to the user in the front end. By aggregating the model predictions, the user can see how each model's prediction compares to the other predictions, and gain more confidence in the prediction if multiple models are classifying the audio as the same class. Additionally, weightings are applied to the models based on the confidence given to each model, and an overall classification is presented based on this as shown in Fig. 8.

4.4. RQ4 - Comparison with state-of-the-art approaches

By evaluating the methodologies and approaches in the literature (cf Table 15), we examine the attributes of state-of-the-art approaches and the proposed approach to voice spoofing detection under different approaches. Here are the questions for comparison under each feature.

- **Classification Problem:** Is the voice spoofing detection approach formulated as a binary or a multiclass problem?

- **Feature Evaluation:** Did the technique involve the comparative analysis of single or multiple features?
- **Methodology:** Does the approach consider a future extraction technique or use the raw audio files as inputs to the solution?
- **Feature Extraction:** Were the features extracted using hand-crafted features, deep feature extraction, librosa spectral features etc.?
- **Deep Learning Algorithms:** Were multiple deep learning algorithms investigated for suitability?
- **Evaluation Dataset:** Did the evaluation go beyond the pre-recorded dataset?
- **Performance Metrics:** What were the performance metrics used to evaluate the model — precision–recall, sensitivity–specificity metrics, EER (Equal error rate), tandem detection cost function (t-DCF) etc.?

Our approach is the first to consider the multiclass problem of voice spoofing detection by considering the classification of converted, synthesised and replay voice recordings against bonafide audio. This is also reflected in the datasets used for evaluation and real-time classification with audio captured from a microphone and user-uploaded audio. As shown in Table 15, all but one of the approaches in the literature investigated the voice spoofing detection problem as a binary classification problem which suffers generalisation to unseen spoofing attacks in the testing stage. Even when the detection of voice spoofing detection is viewed as a multiclass problem, such as in Baumann et al. (2021), only one attack – in this case, the identification of first and second-order replays versus genuine audio – is taken into account. Furthermore, the proposed approach investigated a variety of deep learning models for the classification problems, making use of three classification approaches (image classification of audio data graphs in CNNs, classification of raw audio in WaveNet, and classification of audio features extracted as float values in GRUs and LSTMs), as

Table 15

Comparing state-of-the-art approaches with the proposed approach (Alzantot et al., 2019; Arif et al., 2021; Ballesteros et al., 2021; Baumann et al., 2021; Chen et al., 2017; Delgado et al., 2018; Gomez-Alanis et al., 2019; Gong et al., 2019; Hassan & Javed, 2021; Zhang et al., 2021; Zhou et al., 2022).

Feature	Components	References											
		Baumann et al. (2021)	Delgado et al. (2018)	Gong et al. (2019)	Ballesteros et al. (2021)	Hassan and Javed (2021)	Zhou et al. (2022)	Zhang et al. (2021)	Chen et al. (2017)	Arif et al. (2021)	Gomez-Alanis et al. (2019)	Alzantot et al. (2019)	Ours
Classification problem	Binary	●	●	●	●	●	●	●	●	●	●	●	●
	Multiclass	○											●
Feature evaluation	Single	●		●									
	Multiple		○		○	○	○		○	○	○	○	●
Methodology	Feature extraction	●	●	●	●	●	●	●	●	●	●	●	●
	Raw audio												●
Feature extraction	Librosa spectral features												●
	Deep features										●		
	Other features	●	●	●	●	●	●	●	●	●			
Deep learning algorithms	Single	●	●	●	●	●	●	●		●		●	
	Multiple								●		●		●
Algorithm	CNN				●			●	●		●	●	●
	LSTM					●	●			●		●	●
	GMM	●	●	●					●				●
	GRU										●		●
	WaveNet												●
Evaluation dataset	Pre-recorded	●	●	●	●	●	●	●	●	●	●		●
	Novel Audio												●
	User Input												●
Performance metric	Precision-Recall				●	●	●						●
	Sensitivity - Specificity												●
	EER	●	●	●			●	●	●	●	●	●	
	t-DCF									●	●	●	●
	Other thresholds												●

●: Full coverage ○: Partial coverage ○: No coverage.

opposed to state-of-the-art approaches which mostly investigate a single approach to classification such as Gaussian Mixture Models or LSTMs. As a result, we validated the use of WaveNet, a model designed for use in text-to-speech and music applications, as a suitable model for multiclass classification problems in voice spoofing detection. With a false positive rate of 0.0260, we demonstrate the effectiveness of WaveNet in classifying replay, synthesised and converted attacks against bonafide audio. To our knowledge, we were the first to introduce WaveNet as a suitable algorithm for addressing voice spoofing countermeasures.

Furthermore, this paper investigates the significance of 13 librosa spectral features to algorithm performance, compared with state-of-the-art approaches which mostly make use of one or two features. The most popular ASV features for voice spoofing detection are the Gammatone frequency cepstral coefficients (GFCCs) based on Gammatone filters (Adiga, Magimai, & Seelamantula, 2013; Hanilci, Kinnunen, Sahidullah, & Sizov, 2016), Linear Frequency Cepstral Coefficients (LFCC) (Sahidullah, Kinnunen, & Hanilci, 2015), MFCC (Korshunov et al., 2016), Constant-Q Cepstral Coefficients (CQCC) (Todisco, Delgado, & Evans, 2016) and the perceptual linear prediction coefficients (PLP) (Hermansky, 1990), with the most promising being the Constant-Q Cepstral Coefficients (CQCCs). The proposed approach expands on the state-of-the-art approaches and provides a further contribution to the significance of both widely used features such as the CQCC and MFCC and novel features to the voice spoofing classification problem. To our knowledge, we were the first to introduce and validate librosa spectral features as a reliable feature extraction method for addressing voice spoofing countermeasures. With our approach, we have demonstrated the validity of Librosa's spectral features such as

Spectral Centroid, MelSpectrogram, and Spectral Contrast as promising features for voice spoofing detection. Also by using the raw audio as inputs in the WaveNet architecture, we have demonstrated that using a generative model of raw audio waveforms trained on speech samples is suitable for voice spoofing detection. While librosa is for music and audio analysis for music information retrieval systems, their utility in feature extraction for voice spoofing detection is promising. Although it is challenging to directly compare the proposed model performance to state-of-the-art due to differences in algorithms, datasets, classification problems, features, and evaluative metrics, we do so in some cases where the same algorithms have been used. In Ballesteros et al. (2021), which focuses on the binary classification of real and synthesised voice using CNNs, achieving an F1 score of 0.997, our CNN model for the binary and multiclass classification problem achieved an F1-score of 0.9916 and 0.9917 respectively. Additionally, in Zhou et al. (2022) which uses LSTMs, an F1 score of 0.9484 was achieved in the binary classification of replay audio, while in our approach, the optimal LSTM model for the multiclass classification of replay, synthesised and converted audio against bonafide audio achieved an F1 score of 0.9148. Our approach also takes the evaluation of the models one step further than other implementations, for example rather than only evaluating performance on an evaluation dataset, this paper also evaluates model performance on novel audio data, providing research into the feasibility of real-life use cases for the classification models. Additionally, by deploying the models to a website, we can draw practical insights into the implication of the proposed approach in real time.

5. Conclusion

This paper contributes to the creation and improvement of systems such as automated speaker verification. The voice spoofing detection was formulated as a binary and multiclass classification problem. For the binary classification, the goal is to detect spoofed audio against bonafide audio, while the task is to detect replay, synthesised and converted audio against real audio. The proposed approach makes use of three deep learning models i.e., classification of audio data graphs in CNNs, classification of raw audio waveforms in WaveNet, and classification over the audio features in GRUs and LSTM. The experimental results on the collated dataset, real-time classification using audio captured from a microphone and the deployed model show that the proposed method of voice spoofing detection is effective in learning effective features to distinguish voice spoofing attacks against bonafide audio. This paper found that the spectral centroid was best for CNNs, and this may be because the centroid line was superimposed on top of a spectrogram. This could potentially give the CNN more context, and with multiple lines for each of the features on a single image, it could discern more detail from the graph and classify more accurately. While, GRU and LSTM particularly struggled with the classification of real audio, perhaps training on raw audio may help these models. In the future, we plan to combine the raw waveform and spectral features to jointly construct a hybrid model for the multiclass classification task of voice spoofing detection.

CRedit authorship contribution statement

Jason Boyd: Writing – original draft, Software, Data curation, Investigation, Methodology. **Muhammad Fahim:** Methodology, Validation, Writing – review & editing, Formal analysis. **Oluwafemi Olukoya:** Conceptualization, Methodology, Resources, Writing – review & editing, Data curation, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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